

University of Waterloo Winter 2026 course offerings

The University of Waterloo publishes most course offerings through Quest and the **Schedule of Classes**. Many departments also publish term-specific lists of courses. The tables below summarise the Winter 2026 (term **1261**) offerings that were publicly listed by departments as of early March 2026. Because Waterloo offers hundreds of undergraduate and graduate courses, only departments that publish detailed Winter-2026 lists are covered here. Students should consult the **Schedule of Classes** and **Quest** for complete scheduling (times, locations and instructors).

Cheriton School of Computer Science (graduate courses)

The Cheriton School of Computer Science listed a broad set of graduate courses for Winter 2026. Courses are grouped by research area. See the school's course-offering page for delivery method and prerequisites

1 . 2

Course number	Area / topic (short description)	Notes
CS 636	<i>Intro to Networks & Distributed Systems</i> – available for Master of Health Informatics students ³	For MHI students only
CS 642	<i>Principles of Programming Languages</i> ⁴	Programming Languages area
CS 644	<i>Compiler Construction</i> ⁵	Programming Languages
CS 645	<i>Software Requirement Specification & Analysis</i> ⁶	Software Engineering
CS 646	<i>Software Design, Architectures</i> ⁷	Software Engineering
CS 650	<i>Computer Architecture</i> ⁸	Hardware & Software Systems
CS 651	<i>Data-Intensive Distributed Computing</i> ⁹	Hardware & Software Systems
CS 652	<i>Real-Time Programming</i> ¹⁰	Hardware & Software Systems
CS 654	<i>Distributed Systems</i> ¹⁰	Hardware & Software Systems
CS 656	<i>Computer Networks</i> ¹¹	Hardware & Software Systems

Course number	Area / topic (short description)	Notes
CS 676	<i>Numeric Computation for Financial Modelling</i> ¹²	Scientific & Symbolic Computing
CS 679	<i>Neural Networks</i> ¹³	Scientific & Symbolic Computing
CS 680	<i>Intro to Machine Learning</i> ¹⁴	Computational Statistics
CS 682	<i>Computational Techniques in Biological Sequence Analysis</i> ¹⁵	Bioinformatics
CS 684	<i>Computational Vision</i> ¹⁶	Artificial Intelligence
CS 686	<i>Intro to Artificial Intelligence</i> ¹⁷	Artificial Intelligence
CS 687	<i>Intro to Symbolic Computation</i> ¹⁸	Scientific & Symbolic Computing
CS 688	<i>Intro to Computer Graphics</i> ¹⁹	Graphics & User Interfaces
CS 698	Introductory Research Topics (several sections) – Secure Programming; Intro to Natural Language Processing; Software & Systems Security; Privacy, Crypto & Data Security ²⁰	Multiple lecture sections covering research-topic samplers
CS 738	<i>Data Engineering for Data Science</i> ²¹	Available only for Data-Science students
CS 747	<i>Software Verification Using Proof Assistants</i> ²²	Open background – Programming Languages
CS 764	<i>Computational Complexity</i> ²³	Algorithms & Complexity
CS 778	<i>Numerical Solution of Partial Differential Equations</i> ²⁴	Scientific & Symbolic Computing
CS 792	<i>Data Structures and Standards in Health Informatics</i> ²⁵	Health Informatics
CS 798	Advanced Research Topics – <i>Multicore Programming</i> (LEC 001) and <i>Advanced Topics in Machine Learning Theory</i> (LEC 002) ²⁶	Requires recommended background
CS 846	Advanced Topics in Software Engineering – <i>Advanced Topics in Software Testing and Debugging</i> (SEM 001) and <i>Large Language Models for Software Engineering</i> (SEM 002) ²⁷	

Course number	Area / topic (short description)	Notes
CS 856 SEM 001	<i>Adv Topics in Distributed Computing: Network Transport Protocols: Modeling, Analysis and Performance Exploration</i> ²⁸	
CS 858 SEM 001	<i>Adv Topics in Cryptography, Security and Privacy: Topics in Secure Computation</i> ²⁹	
CS 860 SEM 001	<i>Adv Topics in Algorithms & Complexity: Space Bounded Computation</i> ³⁰	
CS 860 SEM 002	<i>Adv Topics in Algorithms & Complexity: Algorithms for Private Data Analysis</i> ³¹	
CS 888 SEM 001	<i>Adv Topics in Computer Graphics: Advanced Topics in Light Transport</i> ³²	
CS 889 SEM 001	<i>Adv Topics in HCI: Information Visualization</i> ³³	
CS 889 SEM 002	<i>Adv Topics in HCI: Interfaces for Human-AI Interaction</i> ³⁴	
COGSCI 600	<i>Seminar in Cognitive Science</i> ³⁵	Cross-listed with cognitive science

Chemistry (undergraduate courses)

The Department of Chemistry's Winter 2026 course list covers first-, second-, third- and fourth-year courses. Each entry below links to the official calendar description and departmental information ³⁶ . ³⁷

First-year courses

Course	Title and notes
CHEM 120	<i>Physical and Chemical Properties of Matter</i> ³⁸
CHEM 123	<i>General Chemistry 2</i> ³⁹
CHEM 123L	<i>General Chemistry Laboratory 2</i> ⁴⁰
CHEM 125	<i>Chemical Reactions, Equilibria and Kinetics</i> ⁴¹
CHEM 125L	<i>Chemical Reaction Laboratory 2</i> ⁴²
CHEM 140	<i>Introduction to Scientific Calculations</i> ⁴³

Second-year courses

Course	Title
CHEM 212	<i>Structure and Bonding</i> ⁴⁴

Course	Title
CHEM 221	<i>Multi-Component Analysis</i> ⁴⁵
CHEM 224L	<i>Analytical Instrumentation Laboratory</i> ⁴⁶
CHEM 233	<i>Fundamentals of Biochemistry</i> ⁴⁷
CHEM 233L	<i>Fundamentals of Biochemistry Laboratory</i> ⁴⁸
CHEM 237	<i>Introductory Biochemistry</i> ⁴⁹
CHEM 237L	<i>Introductory Biochemistry Laboratory</i> ⁵⁰
CHEM 254	<i>Introductory Chemical Thermodynamics</i> ⁵¹
CHEM 262	<i>Organic Chemistry for Engineering</i> ⁵²
CHEM 262L	<i>Organic Chemistry Laboratory for Engineering Students</i> ⁵³
CHEM 265	<i>Organic Chemistry 2</i> ⁵⁴
CHEM 265L	<i>Organic Chemistry Laboratory 1</i> ⁵⁵
CHEM 267	<i>Basic Organic Chemistry 2</i> ⁵⁶
CHEM 267L	<i>Organic Chemistry Laboratory II</i> ⁵⁷

Third-year courses

Course	Title
CHEM 313	<i>Main Group and Solid-State Chemistry</i> ⁵⁸
CHEM 313L	<i>Inorganic Chemistry Laboratory 1</i> ⁵⁹
CHEM 335L	<i>Advanced Biochemistry Laboratory</i> ⁶⁰
CHEM 340	<i>Introductory Computational Chemistry</i> ⁶¹
CHEM 350	<i>Chemical Kinetics</i> ⁶²
CHEM 357	<i>Physical Biochemistry</i> ⁶³
CHEM 360	<i>Organic Chemistry 3</i> ⁶⁴
CHEM 360L	<i>Senior Organic Chemistry Laboratory</i> ⁶⁵
CHEM 363	<i>Organic Process Chemistry</i> ⁶⁶
CHEM 370	<i>Introduction to Polymer Science</i> ⁶⁷
CHEM 383	<i>Medicinal Chemistry</i> ⁶⁸
CHEM 392A & 392B	<i>Research Project 1 & 2 for exchange students</i> ⁶⁹

Fourth-year courses

Course	Notes
CHEM 400	Several special topics: <i>Electrochemistry; Chemical Spectroscopy for Biomedical Applications; Transitional Metal Catalysis in Organic Chemistry; Computational Chemistry in the Age of AI; Polymerizations beyond Radical and Condensation Polymerization</i> ⁷⁰
CHEM 430	<i>Antibiotics and Resistance</i> ⁷¹
CHEM 433	<i>Advanced Biochemistry</i> ⁷²
CHEM 494A & 494B	<i>Research Project</i> ⁷³
CHEM 496	<i>Advanced Research Project</i> (exchange-students only) ⁷⁴

Electrical & Computer Engineering (ECE) graduate courses

The ECE department's Winter 2026 page lists graduate courses with tentative instructors ⁷⁵. The following table summarises these offerings.

Course	Title / topic	Instructor (if listed)
ECE 602	<i>Introduction to Optimization</i> ⁷⁶	O. Michailovich
ECE 610	<i>Broadband Communication Networks</i> ⁷⁷	A. Wasef
ECE 612	<i>Information Theory</i> ⁷⁸	P. Mitran
ECE 613	<i>Image Processing and Visual Communication</i> ⁷⁹	O. Michailovich
ECE 627	<i>Register-transfer-level Digital Systems</i> ⁸⁰	A. Boutros
ECE 628	<i>Computer Network Security</i> ⁸¹	G. Gong
ECE 631	<i>Microelectronic Processing Technology</i> ⁸²	W. Wong
ECE 634	<i>Organic Electronics</i> ⁸³	H. Aziz
ECE 638	<i>Biosensing: Fundamentals and Applications</i> ⁸⁴	M. Poudineh
ECE 650	<i>Methods and Tools for Software Engineering</i> ⁸⁵	M. Crowley
ECE 651	<i>Foundations of Software Engineering</i> ⁸⁶	K. Czarnecki
ECE 653	<i>Software Testing, Quality Assurance and Maintenance</i> ⁸⁷	W. Shang
ECE 656	<i>Database Systems</i> ⁸⁸	J. Zhang
ECE 657A	<i>Introduction to Artificial Intelligence</i> ⁸⁹	M. Crowley

Course	Title / topic	Instructor (if listed)
ECE 657D	<i>Neural Networks and Deep Learning</i> ⁹⁰	E. Creager
ECE 659	<i>Intelligent Sensors and Wireless Sensor Networks</i> ⁹¹	O. Basir
ECE 666	<i>Power Systems Operation</i> ⁹²	K. Bhattacharya
ECE 668	<i>Distribution System Engineering</i> ⁹³	R. El-Shatshat
ECE 672	<i>Optoelectronic Devices</i> ⁹⁴	S. Saini
ECE 676	<i>Quantum Information Processing Devices</i> ⁹⁵	B. Hauer
ECE 676C	<i>Programming of Quantum Computing Algorithms</i> ⁹⁶	N.Y. Kim
ECE 677	<i>Applied Quantum Mechanics</i> ⁹⁷	A. Majedi
ECE 699A/B	<i>MEng Projects</i> ⁹⁸	L.L. Xie
ECE 700 Topic 09	<i>Special Topics – Ultrasound in Medicine and Biology</i> ⁹⁹	D. Xiao
ECE 700 Topic 10	<i>Special Topics – Distributed Optimization</i> ¹⁰⁰	M. Fisher
ECE 730 Topic 25	<i>Topics in Solid State Devices – Microfluidic and Nanobitech Systems</i> ¹⁰¹	M. Poudineh
ECE 752	<i>Foundations of Multi-Agent Systems</i> ¹⁰²	S. Zahedi
ECE 757A	<i>Embodied Intelligence</i> ¹⁰³	K. Dautenhahn
ECE 765	<i>Power System Protection and Relaying</i> ¹⁰⁴	S. Azad
ECE 774	<i>Radio and Wireless Systems</i> ¹⁰⁵	S. Boumaiza
ECE 787	<i>Social Robotics</i> ¹⁰⁶	K. Dautenhahn
ECE 6614PD	<i>Industrial Utilization of Electrical Energy</i> (online power-MEng) ¹⁰⁷	R. El-Shatshat
ECE 6616PD	<i>Electric Safety and Grounding System Design</i> (online power-MEng) ¹⁰⁷	M. Salama

Economics (graduate courses)

The Department of Economics published Winter 2026 graduate course descriptions. These courses typically carry 600- and 700-level numbers ¹⁰⁸ . ¹⁰⁹ The list below summarises them.

Course	Focus
ECON 602	<i>Macroeconomic Theory I</i> – provides methodological tools for dynamic equilibrium analysis ¹¹⁰

Course	Focus
ECON 623	<i>Time Series Econometrics</i> – fundamental and modern time-series models and methods ¹¹¹
ECON 626	<i>Machine Learning for Economists</i> – predictive methods and exploratory analysis applied to economic data ¹¹²
ECON 641	<i>Public Economics</i> – economic role of the public sector, provision of public goods, taxation and fiscal federalism ¹¹³
ECON 643	<i>Health Economics</i> – economics of health care and policy, health-insurance markets and government role ¹¹⁴
ECON 672	<i>Financial Economics</i> – utility theory, no-arbitrage pricing, asymmetric information and capital-structure theory ¹¹⁵
ECON 673	<i>Strategic decision-making with uncertainty</i> – game-theoretic analysis of screening, moral hazard, auction design and AI-driven market design ¹¹⁶
ECON 701	<i>Micro II</i> – general equilibrium, game theory, mechanism design and bargaining applications ¹¹⁷
ECON 703	<i>Advanced Macroeconomics: Problems and Policies</i> – research-oriented course covering growth, inequality, fiscal policy and labour markets ¹¹⁸
ECON 721	<i>Econometrics II</i> – systems of equations, generalized method of moments and other advanced econometric methods ¹¹⁹
ECON 723	<i>Applied Macroeconometrics II</i> – time-series analysis, VAR, cointegration, neural networks and DSGE modelling ¹²⁰

Religious Studies (RCS) undergraduate courses

Religious Studies lists courses by 100-, 200- and 300-level bands; modes of delivery (online or in-class) are indicated ¹²¹ . ¹²²

Level & course	Description	Mode
RCS 100 – Religions of Asia	Survey of religions in Asia ¹²³	Online
RCS 102 – Introductory Ancient Greek 2 (held with GRK 102)	Second half of introductory ancient Greek ¹²⁴	Online
RCS 110 – Judaism, Christianity, Islam	Comparative introduction to three Abrahamic religions ¹²⁵	Online
RCS 121 – Evil	Exploration of evil in religious thought ¹²⁶	Online
RCS 224 – Selections from Greek Authors (held with GRK 202)	Readings from Greek texts ¹²⁷	In-class

Level & course	Description	Mode
RCS 235 – Jesus: Life and Legacy (held with JS 235)	Study of Jesus' life and influence ¹²⁸	Online
RCS 253 – Christianity's Big Questions	Topics in Christian theology ¹²⁹	In-class
RCS 285 – Spirituality, Religion and Ecology (held with ERS 294)	Intersections of spirituality and ecology ¹³⁰	In-class
RCS 288 – Health, Medicine and Spirituality (held with HHUM 288)	Religion and spirituality in health contexts ¹³¹	In-class
RCS 318 – Indigenous Worldviews and Spirituality (held with INDG 318)	Indigenous spiritual perspectives ¹³²	In-class
RCS 335 – Christianity, Violence and the Bible (held with PACS 336)	Critical examination of violence in Christian scripture ¹³³	In-class
RCS 398 – Directed Readings: Special Subjects	Independent study course ¹³⁴	Mode not specified

(No 400-level RCS courses were listed on the Winter 2026 page.)

Philosophy (graduate courses)

Graduate offerings from the Philosophy department include special-topics seminars at the 600-level. The Winter 2026 list includes three courses ¹³⁵ :

Course	Theme
PHIL 673/675 — 001	<i>Equality, Sufficiency and Limit</i> – examines egalitarian, sufficientarian and limitarian views in political philosophy ¹³⁶
PHIL 673/675 — 002	<i>Fairness and Anti-Discrimination in AI</i> – studies algorithmic discrimination and philosophical approaches to fairness ¹³⁷
PHIL 680 B	<i>Academic Freedom</i> – pro-seminar exploring the history and debates surrounding academic freedom ¹³⁸

Gender and Social Justice (GSJ) undergraduate courses

The Gender and Social Justice program lists cross-listed courses with other departments for Winter 2026 ¹³⁹ .

Course (with cross-listings)	Title / topic	Delivery
GSJ 102	<i>The Global South</i> – introduction to gender and globalization ¹⁴⁰	On-campus

Course (with cross-listings)	Title / topic	Delivery
GSJ 208E / ENGL 208E	<i>Woman's Writing</i> ¹⁴¹	On-campus
GSJ 222 / PHIL 202	<i>Gender Issues</i> ¹⁴²	On-campus
GSJ 260 / HLTH 260	<i>Social Determinants of Health</i> ¹⁴³	On-campus
GSJ 302 / GSJ 304 / PHIL 302	<i>Feminisms for Social Justice</i> – advanced seminar ¹⁴⁴	On-campus
GSJ 309 / COMMST 434 / ENGL 309G / HIST 309	<i>The Discourse of Dissent</i> ¹⁴⁵	On-campus
GSJ 326 / LS 325 / SOC 325	<i>Sexuality and the Law</i> ¹⁴⁶	On-campus
GSJ 347 / HIST 347	<i>Witches, Wives and Whores</i> ¹⁴⁷	Online
GSJ 401 / HLTH 401 / 662	<i>Global Health</i> ¹⁴⁸	On-campus
GSJ 410 / ENGL 410	<i>18th-Century Women Writers</i> ¹⁴⁹	On-campus

Geography & Environmental Management (GEM) graduate courses

Planned Winter 2026 offerings in the Geography program include several graduate courses ¹⁵⁰ .

Course	Title
GEOG 603	<i>Remote Sensing and Earth System Science</i> ¹⁵¹
GEOG 604	<i>Spatial Statistics</i> ¹⁵²
GEOG 613	<i>Innovation & Economic Development in Cities and Regions</i> ¹⁵³
GEOG 617	<i>Applied Statistics in Ecology and Environment</i> ¹⁵⁴
GEOG 625	<i>Qualitative Methods in Geography</i> ¹⁵⁵
GEOG 662	<i>Transforming Canadian Resource Management</i> ¹⁵⁶
GEOG 669	<i>Energy Sustainability</i> ¹⁵⁷
GEOG 675	<i>Selected Topics in Geography (Reading Course)</i> ¹⁵⁸
GEOG 700B	<i>Geographic Scholarship and Practice 2 – Master's</i> ¹⁵⁹
GEOG 800B	<i>Geographic Scholarship and Practice 2 – Doctoral</i> ¹⁶⁰

Pure Mathematics (graduate courses)

The Pure Mathematics department listed Winter 2026 graduate course offerings ¹⁶¹ :

Course	Title
PMATH 667/467	<i>Algebraic Topology</i> ¹⁶²
PMATH 734/434	<i>Set Theory</i> ¹⁶³
PMATH 745/445	<i>Representations of Finite Groups</i> ¹⁶⁴
PMATH 754/454	<i>Fourier Analysis</i> ¹⁶⁵
PMATH 764/464	<i>Algebraic Geometry</i> ¹⁶⁶
PMATH 833	<i>Harmonic Analysis</i> ¹⁶⁷
PMATH 863	<i>Introduction to Lie Groups and Lie Algebras</i> ¹⁶⁸
PMATH 868	<i>Connections and Riemannian Geometry</i> ¹⁶⁹
PMATH 940	<i>Topics in Number Theory: Fermat's Last Theorem</i> ¹⁷⁰
PMATH 950	<i>Topics in Analysis: Choquet Theory</i> ¹⁷¹
PMATH 945	<i>Topics in Algebra: Category Theory and Homological Algebra</i> ¹⁷¹
PMATH 965	<i>Topics in Geometry and Topology: Toric Varieties</i> ¹⁷²

Kinesiology and Health Sciences (undergraduate)

The School of Kinesiology and Health Sciences publishes a three-year teaching plan with term-by-term offerings. The Winter 2026 column indicates which courses will be offered ¹⁷³ ¹⁷⁴ ¹⁷⁵. The list below compiles courses marked as **Offered** or **Online** for Winter 2026.

100-level courses

- **KIN 100** and **KIN 100L** – introduction to human movement and lab ¹⁷⁶ .
- **KIN 104** and **KIN 104L** – human anatomy and physiology with lab ¹⁷⁶ .
- **KIN 121** and **KIN 121L** – mechanics of human movement with lab ¹⁷⁷ .
- **KIN 140L** – introduction to scientific writing laboratory ¹⁷⁸ .
- **KIN 146** – mental health and wellness ¹⁷⁹ .

200-level courses

- **KIN 202** and **KIN 202L** – biomechanics and lab ¹⁸⁰ .
- **KIN 204** and **KIN 204L** – human physiology 2 and lab ¹⁸¹ .
- **KIN 217** – introduction to exercise psychology ¹⁸² .
- **KIN 221** and **KIN 221L** – human motor control and lab ¹⁸³ .
- **KIN 232** – exercise prescription and assessment ¹⁸⁴ .
- **KIN 255** and **KIN 255L** – exercise physiology and lab ¹⁸⁵ .

300-level courses

Winter-2026 offerings include (asterisks indicate courses delivered online):

- **KIN 301, KIN 305, KIN 308, KIN 310, KIN 314, KIN 320, KIN 326, KIN 327, KIN 340, KIN 342, KIN 346, KIN 351, KIN 352, KIN 354, KIN 356, KIN 357, KIN 359, KIN 397** ¹⁸⁶ .
- **KIN 307, KIN 343 and KIN 360**** – offered online or hybrid ¹⁸⁷ .

400-level courses

The Winter 2026 column lists many senior-level offerings ¹⁸⁸ :

- **KIN 400, 402, 403, 404, 406, 407, 415, 416, 418, 420, 422, 425, 427, 428, 429, 446, 453, 455, 459, 470** (sections a and b) ¹⁸⁸ .
- **KIN 491 and KIN 492** are offered, while **KIN 390, 391, 431, 432, 433** and **472** (independent research and projects) are available every term ¹⁸⁹ .

Summary

The Winter 2026 term at the **University of Waterloo** features a large and diverse slate of courses across disciplines. Graduate programs in **computer science, electrical & computer engineering, economics, philosophy, gender and social justice, geography** and **pure mathematics** all posted detailed Winter-term course lists, while undergraduate departments such as **chemistry, religious studies**, and **kinesiology and health sciences** described their offerings publicly. Students seeking a comprehensive schedule—including class times and instructors—should consult the **Schedule of Classes** and **Quest**.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

31 32 33 34 35 Winter 2026 | Cheriton School of Computer Science | University of Waterloo

<https://cs.uwaterloo.ca/current-graduate-students/courses/current-course-offerings/winter-2026>

36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65

66 67 68 69 70 71 72 73 74 Winter 2026 course offerings | Chemistry | University of Waterloo

<https://uwaterloo.ca/chemistry/undergraduate-studies/course-offerings/winter-2026-course-offerings>

75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103

104 105 106 107 Winter 2026 Courses | Electrical and Computer Engineering | University of Waterloo

<https://uwaterloo.ca/electrical-computer-engineering/graduate-studies/current-students/courses/winter-2026-courses>

108 109 110 111 112 113 114 115 116 117 118 119 120 Winter 2026 course offerings | Economics | University of Waterloo

<https://uwaterloo.ca/economics/winter-2026-course-offerings>

121 122 123 124 125 126 127 128 129 130 131 132 133 134 Winter 2026 course offerings | Religious Studies | University of Waterloo

<https://uwaterloo.ca/religious-studies/undergraduate-studies/courses/winter-courses>

135 136 137 138 Graduate Courses | Philosophy | University of Waterloo

<https://uwaterloo.ca/philosophy/graduate/graduate-courses>

139 140 141 142 143 144 145 146 147 148 149 **Courses | Gender and Social Justice | University of Waterloo**
<https://uwaterloo.ca/gender-social-justice/undergraduate/courses>

150 151 152 153 154 155 156 157 158 159 160 **GEM Graduate Courses | Geography and Environmental Management | University of Waterloo**
<https://uwaterloo.ca/geography-environmental-management/graduate/masters-and-phd-geography/gem-graduate-courses>

161 162 163 164 165 166 167 168 169 170 171 172 **Course offerings/descriptions | Pure Mathematics | University of Waterloo**
<https://uwaterloo.ca/pure-mathematics/graduate-studies/current-students/course-offeringsdescriptions>

173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 **Course enrolment and planning | Kinesiology and Health Sciences | University of Waterloo**
<https://uwaterloo.ca/kinesiology-health-sciences/current-undergraduates/course-enrollment-and-planning>